

Chalaka Perera

SOFTWARE ENGINEER

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About Me

Final-year Computer Engineering undergraduate and SRE intern at WSO2, building production software across application, cloud, and infrastructure layers. Experienced in full-stack development, Kubernetes, cloud platforms, automation, and AI-powered systems, with a strong focus on reliability and security. I enjoy solving complex engineering problems and turning them into practical, maintainable solutions.

Education

University of Peradeniya

BSc.Eng(Hons.) in Computer Engineering, GPA: 3.971/4.00, First Class

Peradeniya, Sri Lanka

2022 - Present

Royal College

G.C.E. Advanced Level, Physical Science, 3As | Z-score: 2.0291

Colombo 07, Sri Lanka

2018 - 2020

Experience

WSO2

Site Reliability Engineering Intern, On-site

Colombo, Sri Lanka

Apr 2026 - Present

1. Compliance Evidence Submission Portal | 📄 Repo

Agentic Browser Automation | In Production

- Engineers collected audit evidence manually, capturing screenshots and PDF exports from **GitHub and cloud consoles**.
- Built a **full-stack evidence portal** with a browser agent that signs in and captures the evidence automatically.
- Designed reusable templates enabling a single instruction to collect evidence across all matching resources.
- Conducted **threat modelling** and a security review, which approved the portal for **production use** at WSO2.
- Fixed an access-control flaw that allowed unauthenticated downloads, and enforced **single sign-on** on every request.
- Packaged the tool for **one-command installation** and built the repository's first automated **release pipeline**.
- Wrote **unit and end-to-end tests** for every API endpoint, running against a real database instead of mocks.

Tech: Python, FastAPI, Azure PostgreSQL, Azure Blob Storage, browser-use, Playwright, Azure OpenAI, React, TypeScript, WSO2 Identity Platform, OAuth 2.0, WSO2 Developer Platform, GitHub Actions, pytest

2. SRE Ops Pilot (SREP), Grounded Assistant for the SRE Team

Full-Stack | RAG & LLM Evaluation

- SRE runbooks and operational documentation were spread across systems, making routine questions slow to answer.
- Built a **full-stack assistant** that answers from that documentation and cites the source for every statement.
- Added a verification layer that **prevents fabricated answers**, checking every quote against the source document.
- Engineered document selection to send only relevant content, cutting cost and preventing **runaway API spend**.
- Evaluated candidate language models against a purpose-built test set drawn from the team's real questions.
- Selected the model that answered **63 of 65** correctly, rejected every unanswerable question, and ran **twice as fast**.
- Delivered **single sign-on**, conversation history, per-user cost tracking, an audit log and a configurable spend cap.
- Wrote **unit and end-to-end tests** across backend and frontend, with **threat modelling** now under security review.

Tech: Python, FastAPI, Azure OpenAI, PostgreSQL, Alembic, SSE Streaming, React, TypeScript, react-query, WSO2 Identity Platform, OAuth 2.0, pytest, Vitest

University of Peradeniya

Undergraduate Teaching Assistant, Part-time

Peradeniya, Sri Lanka

Sep 2023 - Mar 2026

CO322 Data Structures & Algorithms | CO225 Software Construction | CO2030 Data Structures & Algorithms II

Open Source Contributions

1. Kubernetes Python Client | 📄 #2690 | #2689 | #2688 | #2687

CNCF | 4 PRs Merged

- Four pull requests merged** into the official Kubernetes client for Python, a CNCF project with **7.6k stars**.
- It could only lock leader election on a **ConfigMap**, which **client-go** had dropped. I shipped the missing **Lease** lock.
- Found it stored **local wall-clock time** labelled as **UTC**, silently breaking leader election against **other Kubernetes clients** on any non-UTC host.
- 155 lines** of runtime code behind **17 unit tests**, reviving an abandoned contribution and crediting its original author.
- Earlier merges: the standard **SPDX** license identifier across all three build scripts, and a repair to the **broken documentation build**.

2. sktime | 📄 #10908

Time-Series ML | 1 PR Merged

- Merged into **sktime**, a **10k-star** time-series ML library, documenting two capabilities its **extension templates** omitted.

Projects

- 1. CloudCare HMS, AWS Well-Architected Infrastructure** |  Repo Personal | Cloud, IaC
 - o Built a production-style AWS environment **entirely in code**, using **nine Terraform stacks** with shared, locked state.
 - o Private database across **two availability zones**, with layered firewall rules in front of it.
 - o Deployed through **GitHub Actions** using **short-lived credentials**, eliminating permanent AWS keys entirely.
 - o Ran it with a monitoring dashboard and **eight alarms** over a self-healing server group.**Tech:** AWS, Terraform, GitHub Actions, OIDC, EC2, ALB, RDS, DynamoDB, Lambda, CloudFront, Secrets Manager, CloudWatch, Docker
- 2. CloudCare HMS, EKS Microservices & Observability** |  Repo Personal | Kubernetes, SRE
 - o Split one large application into **four services** on **Kubernetes**, each with its own **Helm** chart and pipeline.
 - o Shipping one service no longer redeploys the rest. Rollback is one command, since every build carries its commit.
 - o Cut scale-up time from **five minutes to thirty seconds**, and watched the cluster through **Prometheus, Grafana and Loki**.**Tech:** AWS EKS, Kubernetes, Helm, Terraform, Prometheus, Grafana, Loki, ALB Ingress, ECR, GitHub Actions, FastAPI, Docker
- 3. LMS University, Full-Stack Digital Education Platform** |  Repo Personal | Full-Stack
 - o One **Spring Boot** API serving two clients at once, a **React** website and a **React Native** mobile app.
 - o Social feed, live messaging, an admin dashboard, and a job board where alumni hire students.
 - o Moved it from Docker Compose onto **Kubernetes**, keeping data intact across restarts and passwords out of config files.**Tech:** Spring Boot, React, React Native (Expo), MySQL, Docker, Kubernetes, JWT, WebSockets, NGINX
- 4. Explainable AI Driven Zero-Trust Anomaly Detection for Encrypted Traffic** Final Year Research | iPURSE 2026
 - o Detects attacks inside **encrypted network traffic without decrypting it**, preserving privacy end to end.
 - o **Explainable AI** shows the analyst *why* a connection was flagged, not just that it was. Tested on **CIC-IDS-2017**.
 - o Presented before an **external expert panel** at **iPURSE 2026**, and the **abstract** was accepted into the proceedings.**Tech:** Python, Semi-Supervised ML, Explainable AI, CIC-IDS-2017, NumPy, Matplotlib
- 5. Lightify, Smart Home IoT Lighting System** |  Repo |  Web Group | My scope: network & firmware
 - o Designed the **network architecture** and wrote the **ESP32 firmware** the whole system runs on, in C and C++.
 - o A new device joins over **Wi-Fi or Bluetooth** and is configured in place, never reprogrammed.
 - o Reads the occupancy sensors, drives the lights, and carries data and commands to **AWS IoT Core** over **MQTT**.**Tech:** ESP32, C, C++, AWS IoT Core, MQTT, Wi-Fi and Bluetooth Provisioning, Spring Boot, MongoDB
- 6. Artery Resolver, Cardiovascular Risk Analysis** |  Repo Group, with Faculty of Medicine | My scope: app, data, CV model
 - o Automated **Flow-Mediated Dilation**, an early cardiovascular-risk test clinicians were measuring by hand.
 - o The tool reads an **ultrasound video**, finds the artery walls with **OpenCV**, and tracks how far it widens.
 - o Added **blood-flow velocity** analysis, turning each scan into a **summarised report** a doctor can act on.
 - o Built the **Java** desktop app and **MySQL** database behind it, with records scoped to each doctor's own patients.**Tech:** Java, Java Swing, JDBC, MySQL, Python, OpenCV, Computer Vision, Ultrasound Video Processing

Technical Skills

- o **Languages:** Python, Java, TypeScript, JavaScript, C, C++, SQL
- o **Backend & Web:** FastAPI, Spring Boot, React, React Native (Expo), REST, WebSockets, SSE
- o **Cloud & IaC:** AWS, Azure, WSO2 Developer Platform, Terraform, Helm, Docker, Kubernetes (EKS), NGINX
- o **CI/CD & Observability:** GitHub Actions, OIDC Federation, Jenkins, Prometheus, Grafana, Loki, CloudWatch
- o **Data & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Alembic, Azure Blob, MQTT
- o **AI & Agents:** Azure OpenAI, RAG, LLM Eval Harnesses, browser-use, Playwright, LangChain, OpenCV
- o **Security & Tools:** Threat Modeling, Enterprise SSO, OAuth 2.0, Git

Professional Development

- o Terraform, **Top 10% Globally** (TestDome, 2026)
- o Introduction to Kubernetes LFS158 (Linux Foundation, 2026)
- o AWS Cloud Practitioner Essentials (Amazon Web Services, 2026)

Extra-Curricular Activities

- o **EngEx 2025:** presented *Lightify* at the University of Peradeniya 75th Anniversary Engineering Exhibition
- o **Student Member**, IESL (2025 - Present) | **Member**, ACES, UoP (2023 - Present) | **Member**, AIESEC (2022 - 2024)
- o **Maths & Science Tutor**, Arunella Education Welfare Programme, student-led seminars for rural schools Jul - Dec 2023

References

Dr. Suneth Namal Karunarathna

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University of Peradeniya, Sri Lanka | namal@eng.pdn.ac.lk

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